

CS & IT ENGINEERING



Operating System

Process Management

Lecture – 1

By– Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Operating System Definition

Topic

Types of Operating System

doubt ?

Topics to be Covered



Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

Topic

Process Control Block



Topic : System Call



A system call is a way for programs to interact with the operating system

mode bit $\Rightarrow$





Topic : Dual Mode of Operation

2 modes:

User Mode (mode bit = 1)

Kernel/System/Supervisor/Privileged Mode (mode bit = 0)

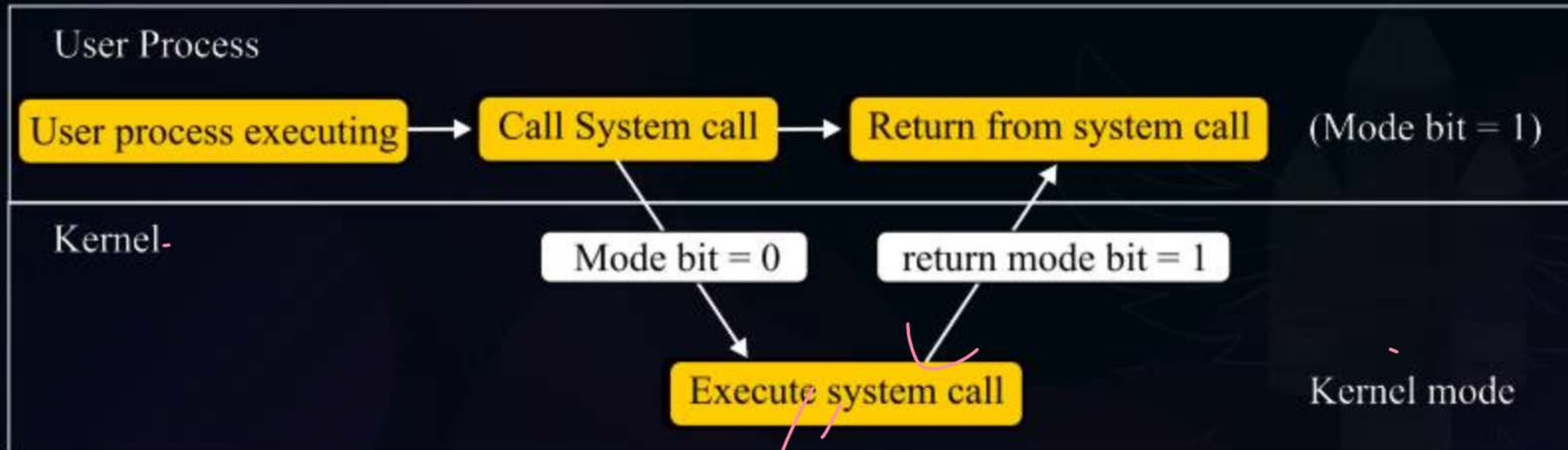


Topic : Dual Mode of Operation

2 Modes:

User Mode (mode bit = 1)

Kernel/System/Supervisor/Privileged Mode (mode bit = 0)





Topic : Process



Process = Program + Runtime activities



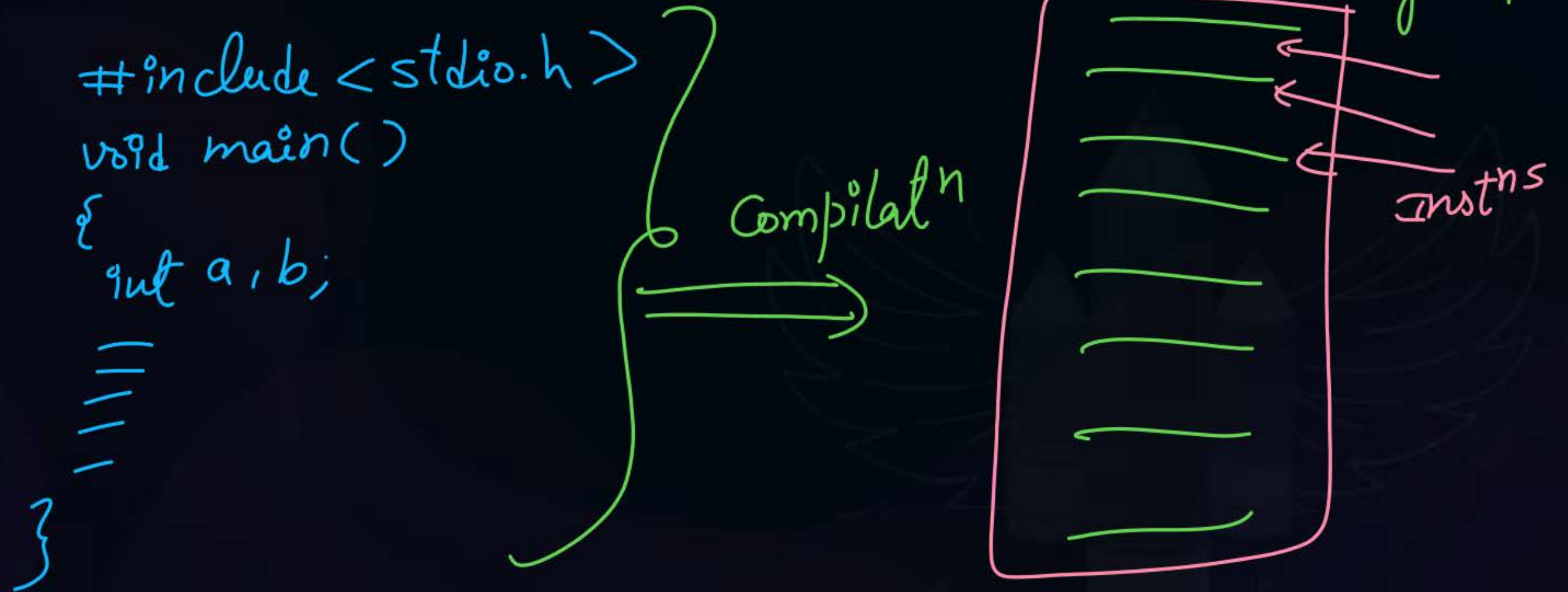


Topic : Process

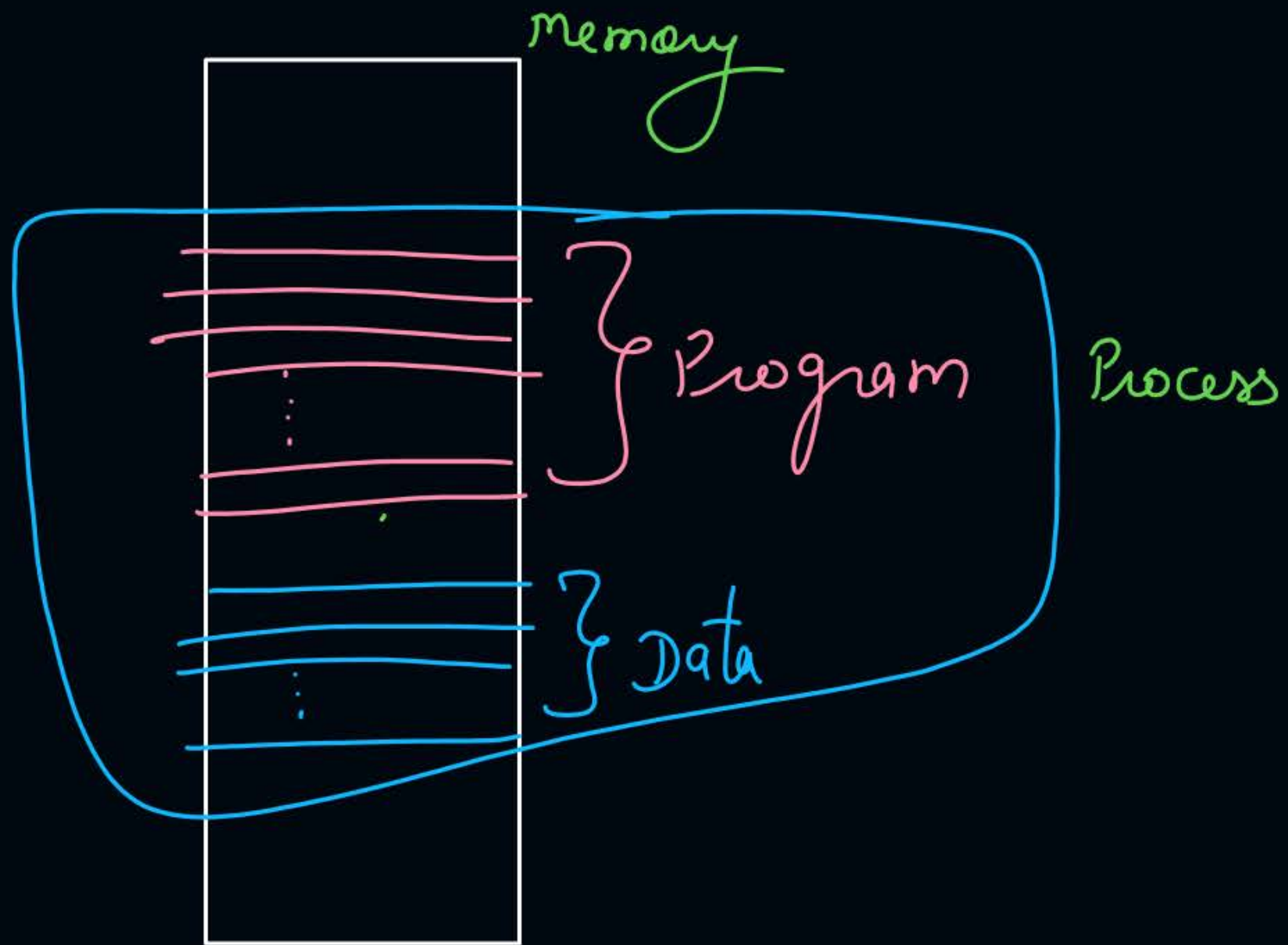


□ Process:

- Program under execution



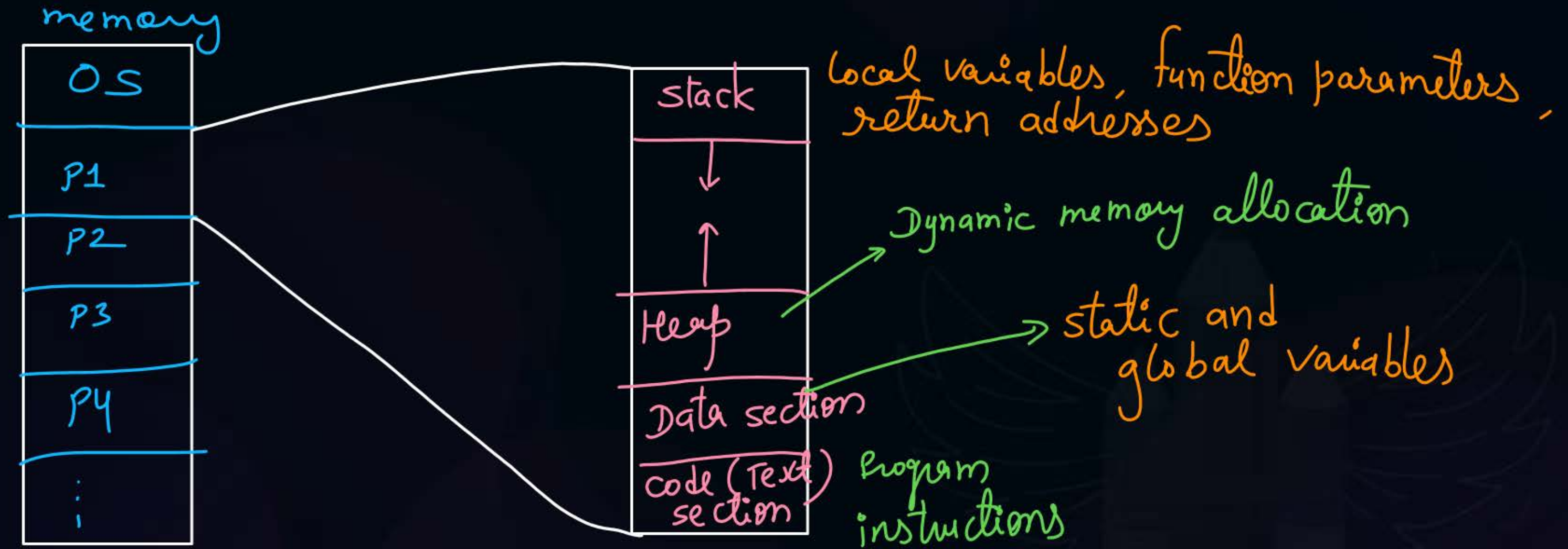
when we try to run program





Topic : Representation of a Process

each process is stored in memory in 4 sections





Topic : Operations on a Process



- Create (Resource Allocation)
- Schedule, Run
- Wait/Block
- Suspend, Resume
- Terminate (Resource Deallocation)

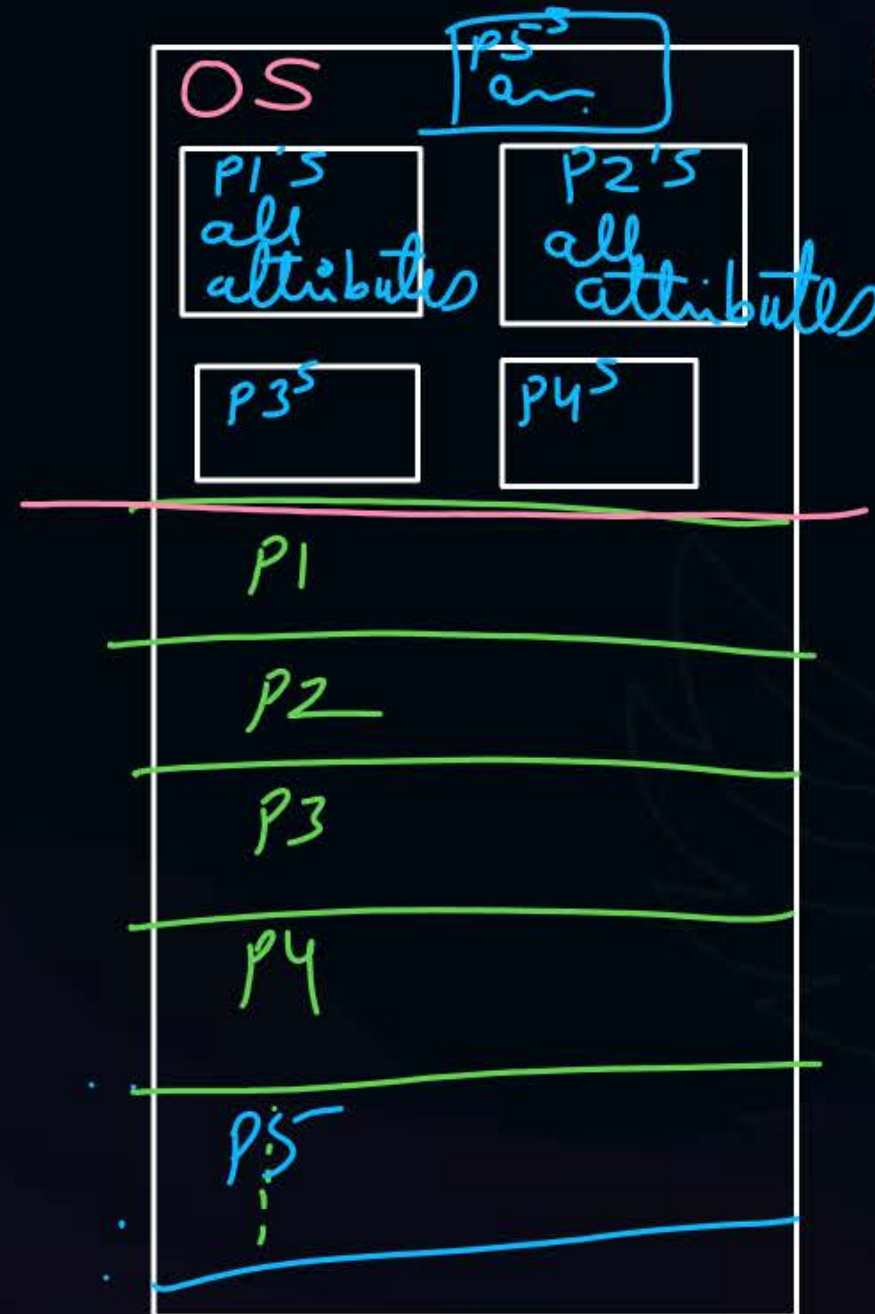




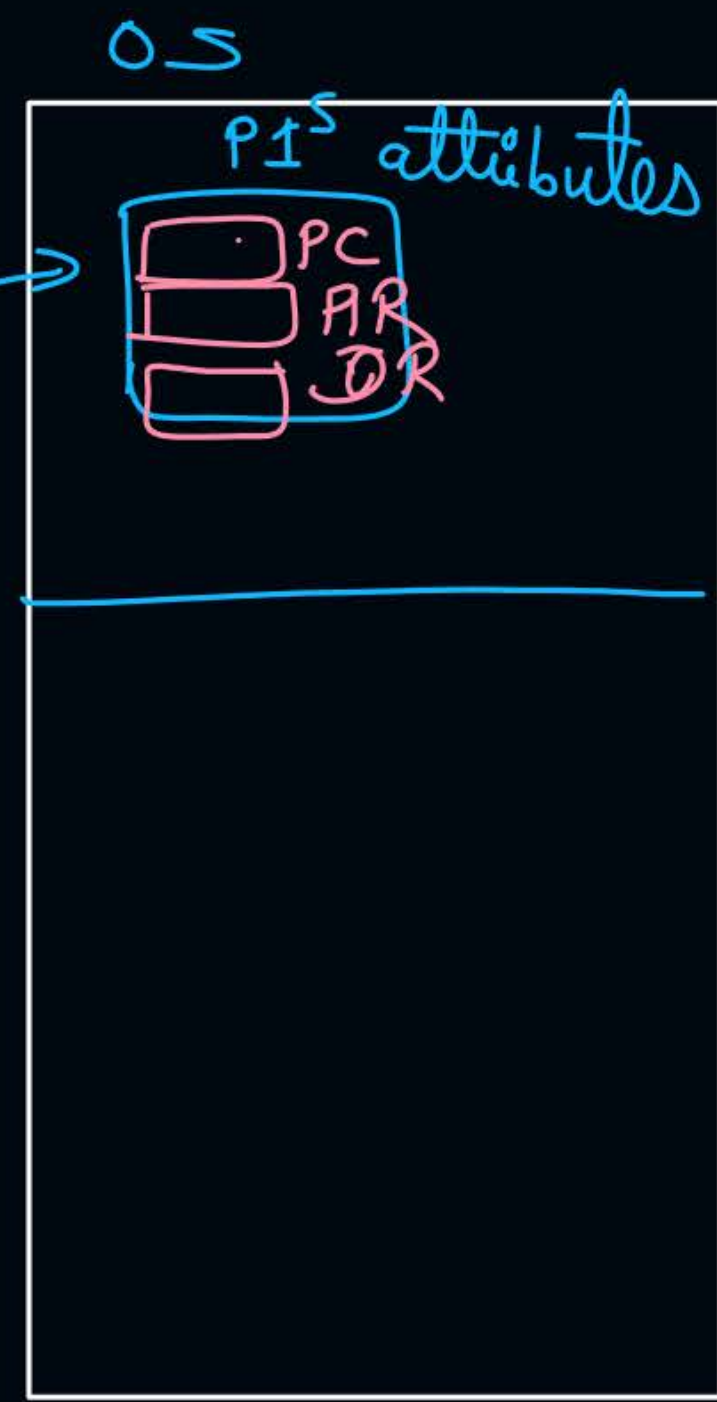
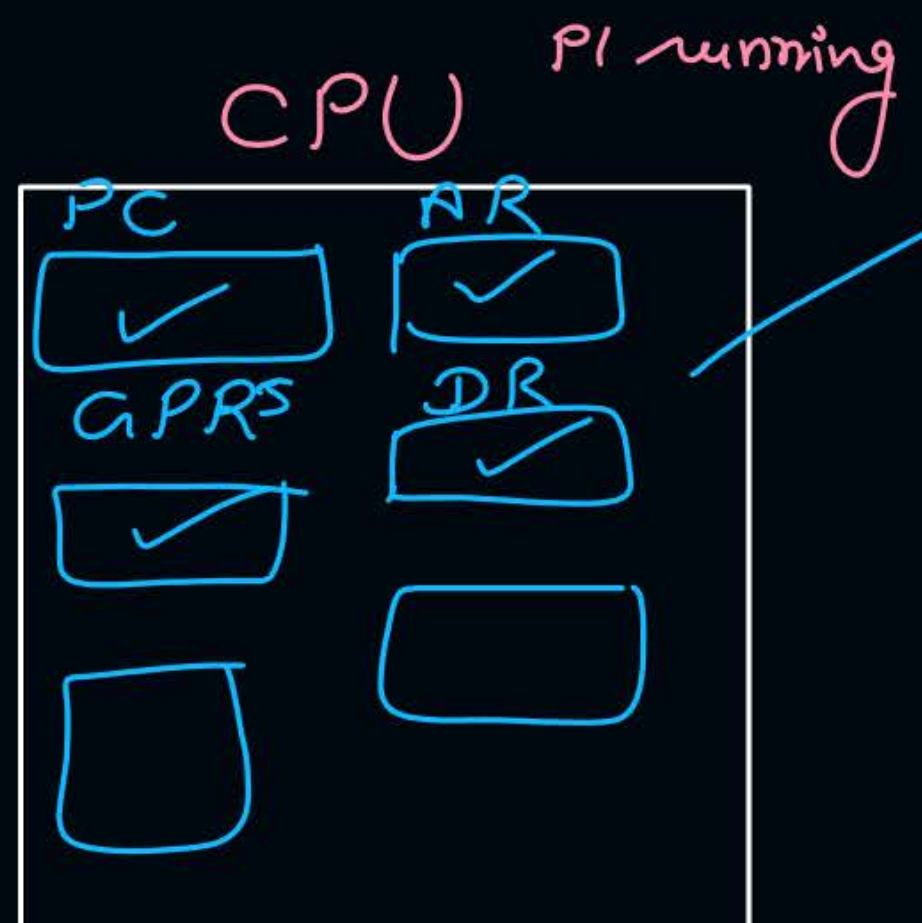
Topic : Attributes of a Process

→ OS maintains attributes about each process.

- ✓ PID (Process id)
- ✓ PC
- ✓ GPR, all other register
- List of Devices
- Type
- Size
- Memory Limits
- Priority
- State
- List of Files



Main memory





Topic : PCB

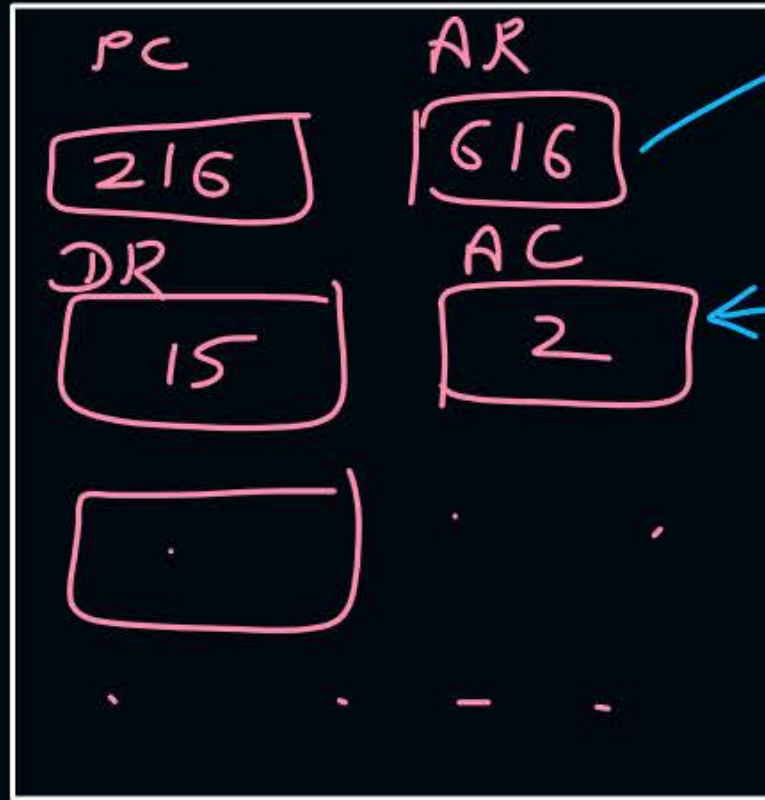


→ Process control block

Also known as process descriptor



CPU P1 in CPU P1





Topic : Context



The content of PCB of a process are collectively know as 'Context' of that process

Context switch :-

- saving context of current running process from CPU to it's PCB in memory
- loading next process's context from it's PCB to CPU.

Dispatcher :-

a program of OS which does context switching.

#Q. While running a process can access its PCB from main memory?

False

[MCQ]

#Q. A process in the context of computing is:

- A** A set of instructions to be executed on a computer → Program
- B** ✓ A program in execution → Process
- C** A piece of hardware that executes a set of instructions → CPU
- D** The main procedure of a program → main()



2 mins Summary

Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

Topic

Process Control Block



Happy Learning

THANK - YOU

